Worksheet 4-3 A Rates in Context #1
AP Calculus

Name SOLUTIONS

1. (Calculator Active) The tide removes sand from a beach at a rate modeled by the function R, given by

$$R(t) = 1 + 6\sin\left(\frac{2\pi}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S, given by $S(t) = \frac{18t}{1+4t}$. Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \le t \le 10$. Find the rate at which the total amount of sand on the beach is changing at time $t = 6$.

Rate at which total amt. of sand on beach is changing $= S(t) - R(t)$

$S(6) - R(6) = -2.668$ yd^3/hr

At $t=6$, the total amount of sand on the beach is decreasing at a rate of 2.668 yd^3/hr

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2. (Calculator Active) Tourists visiting an island resort contracted a mystery illness over a 45-day period. The health authorities recorded the rate of new cases per day and some of the rates are listed in the table at the right.

t Day	$N(t)$ New cases per day
2	3
6	8
10	15
15	30
25	100
35	50
40	22
45	10

a.) Use the table to estimate $N'(20)$. Show the computations that lead to your answer. Indicate units of measure.

$$N'(20) \approx \frac{N(25)-N(15)}{25-15} = \frac{100-30}{10} = \frac{70}{10}$$

$= 7$ new cases per day per day

b.) After studying the spread of the disease, the health department authorities decided they could model the number of new cases per day with the model $R(t) = \frac{80000e^{-0.2t}}{(1+200\ \ ^{-0.2t})^2}$ for $0 \le t \le 50$ days. The disease is considered eradicated when the number of new cases per day does not exceed 5. Use $R(t)$ to find on what day this will occur.

$R(t) = 5$ when $t = 48.274$ days

The disease will be eradicated on the 49th day.

c.) Find $R'(20)$. Using appropriate units, explain the meaning of your answer in terms of the decay rate of the substance.

$R'(20) = 7.697$ new cases per day per day

$R'(20)$ means the number of new cases per day is increasing at a rate of 7.697 cases per day per day on the 20th day

3. (Calculator Active) The number of gallons, $P(t)$, of a pollutant in a lake changes at the rate $P'(t) = 2 - 4e^{-0.3\sqrt{t}}$ gallons per day, where t is measured in days. There are 100 gallons of pollutant in the lake at time $t = 0$. Is the amount of pollutant increasing at time $t = 7$? Why or why not?

$P'(7) = 0.1025396$

The amount of pollutant is increasing at time $t=7$ because $P'(7) > 0$.

4. Use the data from the table to find an approximation for $R'(40)$. Show the computations that lead to your answer. Indicate units of measure.

t (minutes)	$R(t)$ (gallons per minute)
0	15
20	25
30	45
50	60
60	70
90	75

$$R'(40) \approx \frac{R(50)-R(30)}{50-30} = \frac{60-45}{50-30} = \frac{15}{20}$$

$$= 0.75 \ \frac{\text{gal/min}}{\text{min}}$$

OR 0.75 gal/min^2

5. A store is having a 10-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function E defined by $E(t) = 0.2t^4 - 10t^3 + 50t^2$ for $0 \le t \le 10$. At what rate are shoppers entering the store 4 hours after the start of the sale?

$E'(t) = 0.8t^3 - 30t^2 + 100t$

$E'(4) = -28.8$ shoppers/hour

Shoppers are entering the store at a rate of $-28.8 \ \frac{\text{shoppers}}{\text{hr}}$